Content for Today's Experiment

1- Density measurement

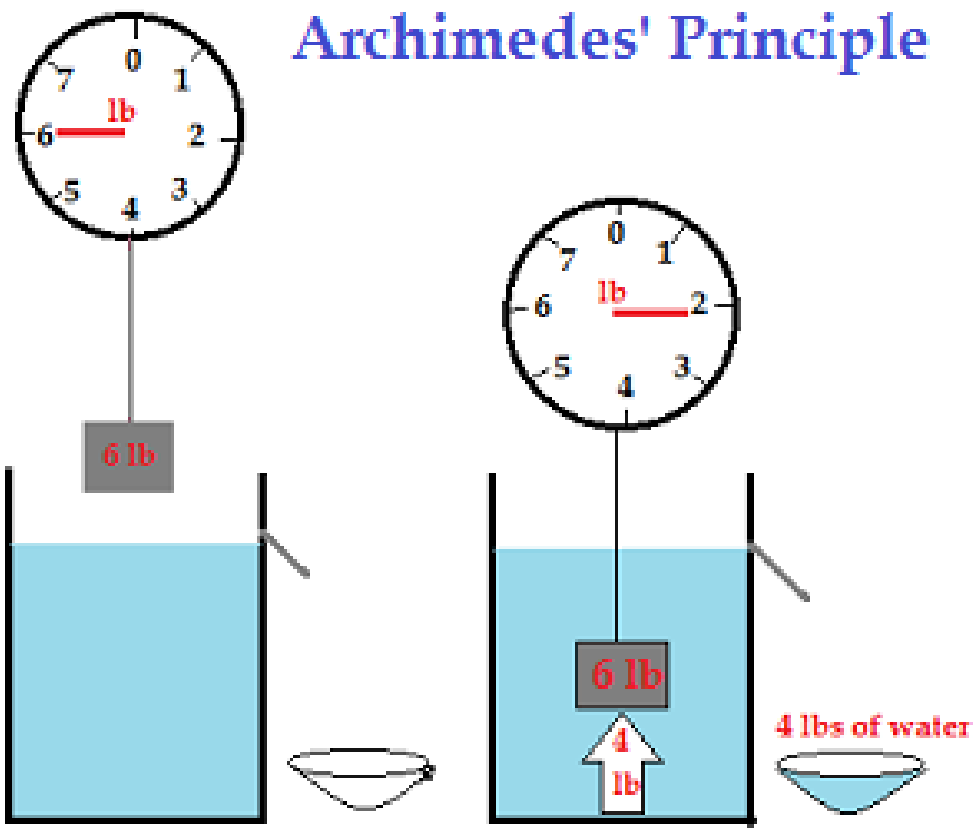
2- Temperature measurement

3- Wear resistance

4- Chemical reactivity

Density measurement Experiment

Archimedes principle- The buoyant force on a submerged object is equal to the weight of the liquid displaced by the object.



$$F_B = \rho g V$$

Density measurement using weighing scale

$$\rho = \frac{A}{A - B}(\rho_0 - \rho_L) + \rho_L$$

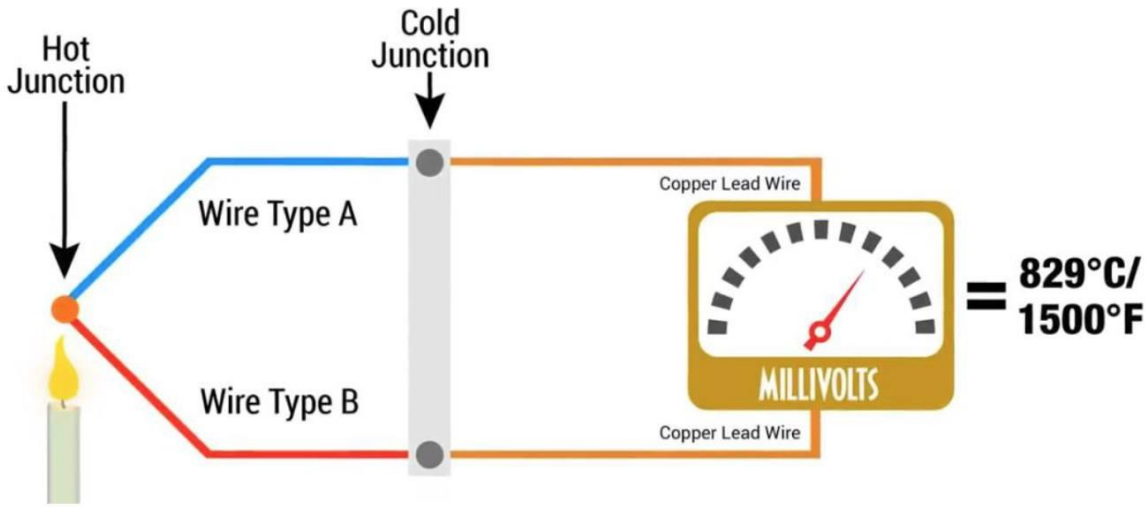
ρ = Density of the sample
A = Weight of the sample in air
B = Weight of the sample in the auxiliary liquid
ρ_0 = Density of the auxiliary liquid
ρ_L = Density of air

Precision balance

It is a balance used to weigh quantities to a very precise number, usually up to one milligram sometimes referred to as “top loading balances”

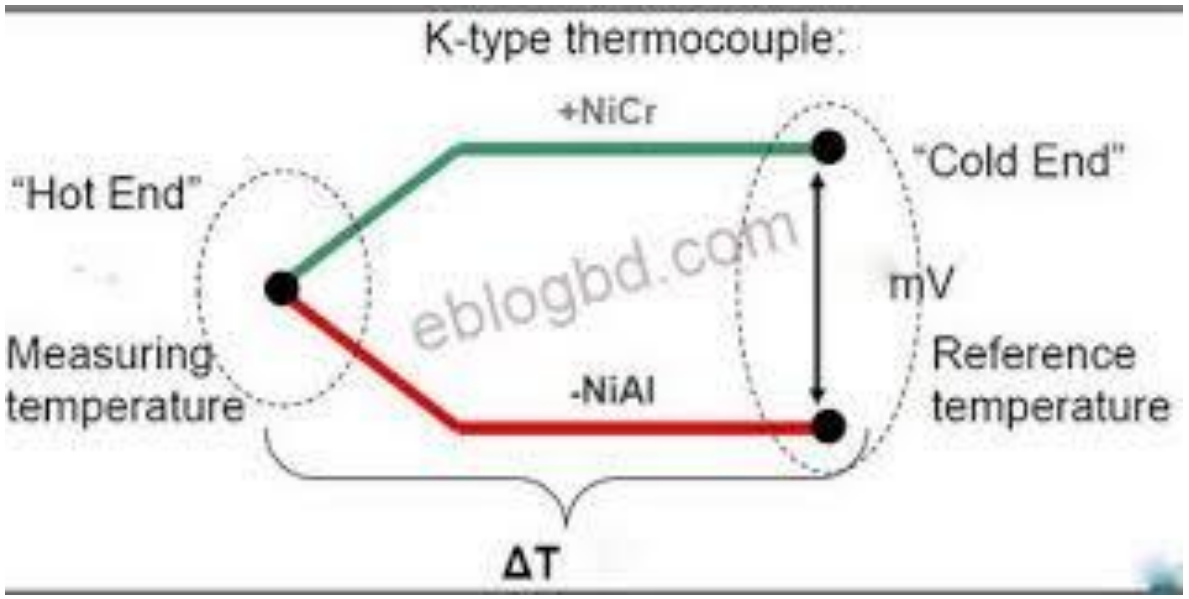
Temperature measurement Experiment

contact type pyrometer with K-type thermocouple



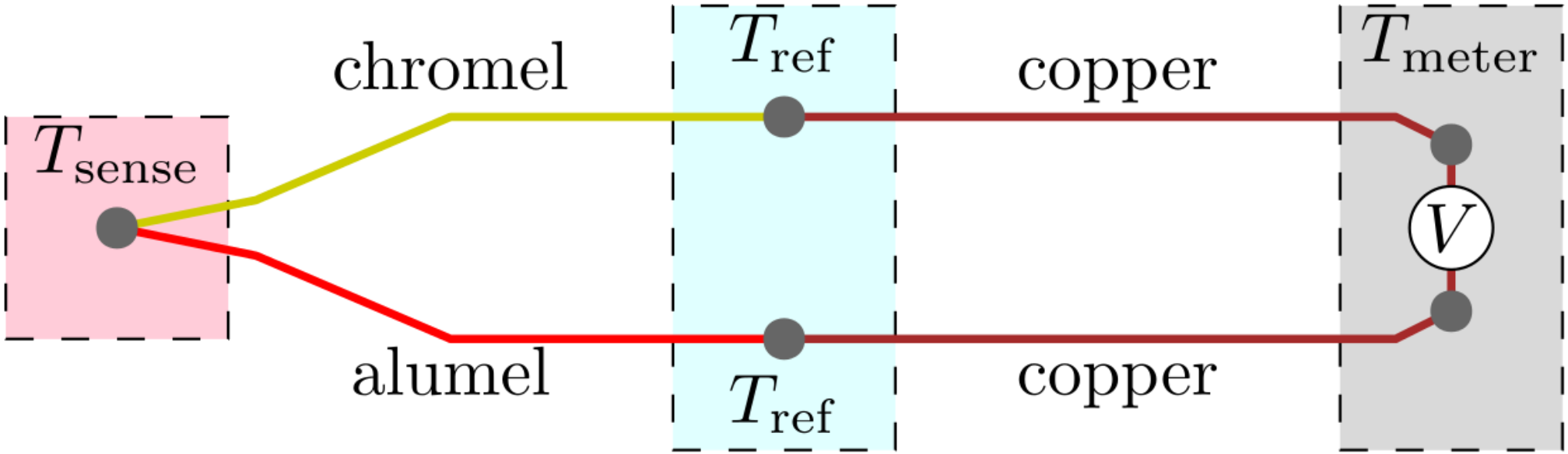
	Overall range	Typical accuracy
Type K (Chromel / Alumel)	-200 to 1300	2.2 °C

Thermocouple working principle



Thermocouples are based on the principle that two wires made of dissimilar materials connected at either end will generate a potential between the two ends that is a function of the materials and temperature difference between the two ends (also called the Seebeck Effect).

K-type thermocouple

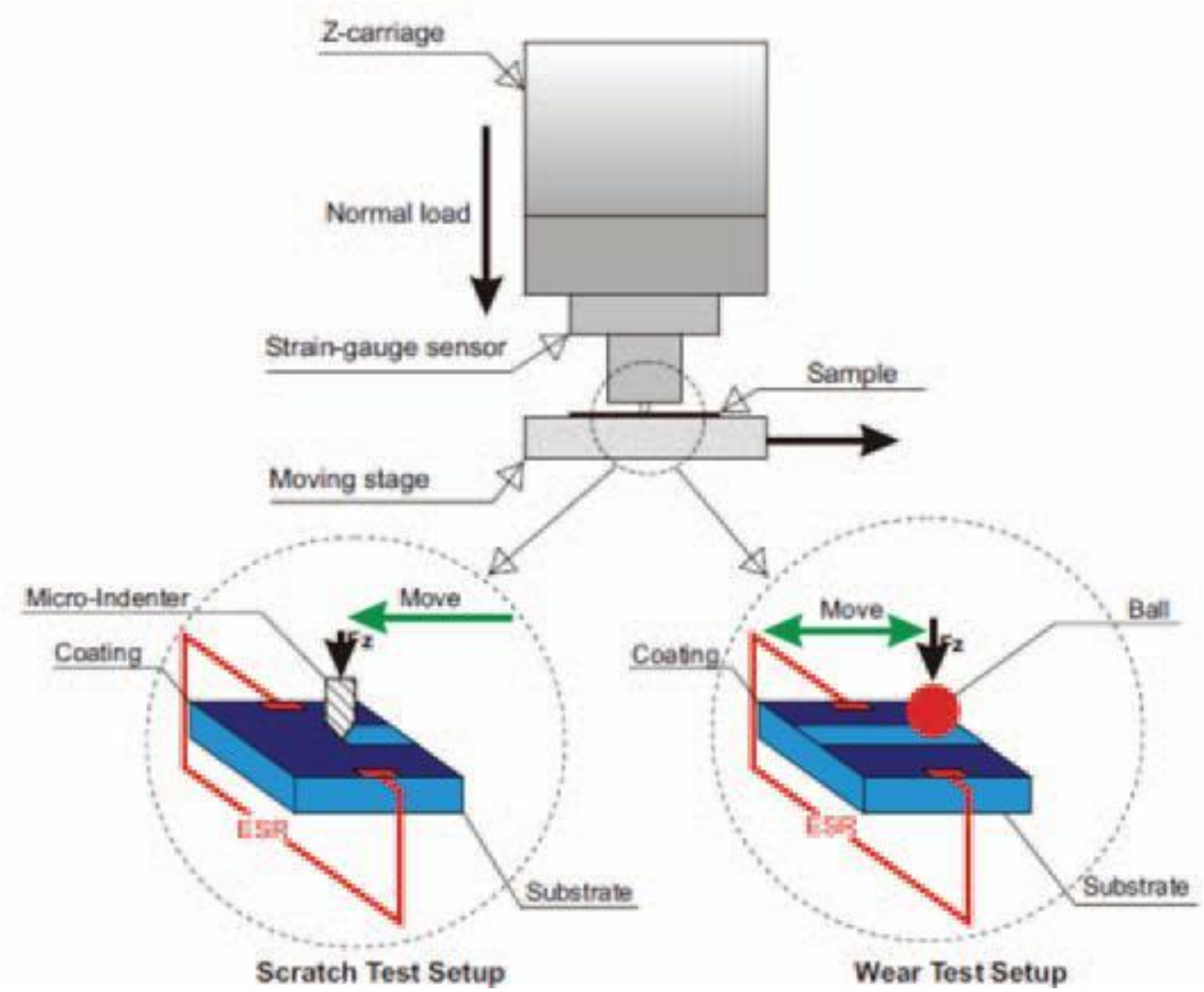


Chromel is an alloy of nickel and chrome plus nine other elements.
Alumel is an alloy containing nickel manganese, aluminum, silicon and nine other elements.
Chromel is the positive wire; **Alumel** is the negative

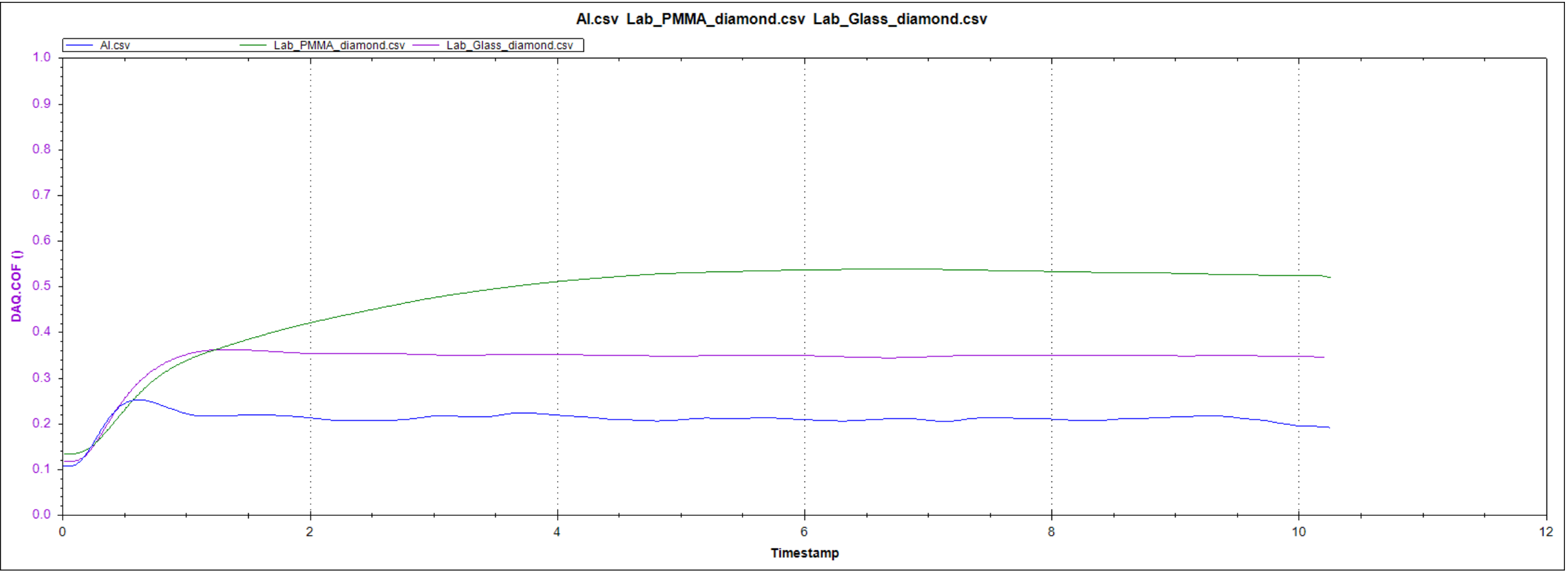
Advantages of thermocouple

- Good accuracy
- Fast thermal response
- Robust; can be used under high vibration and in other harsh environments
- Wide operating temperature range
- Low cost
- Highly reliable

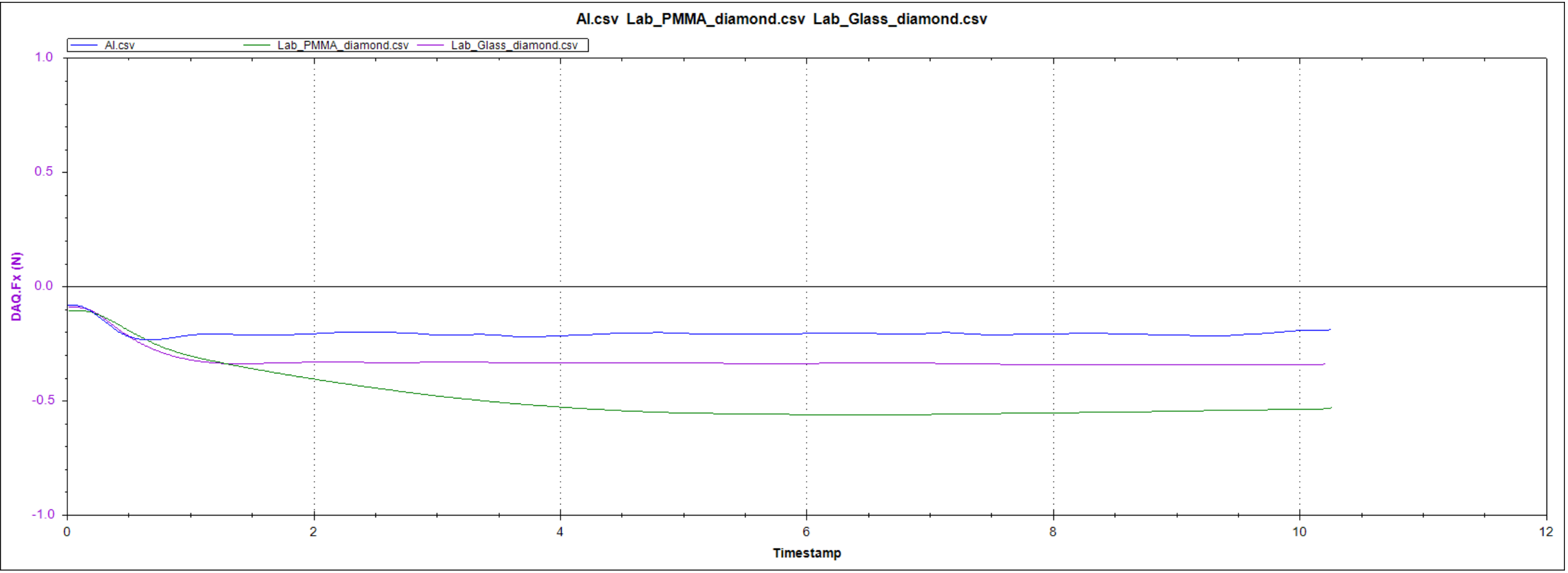
Wear resistance test



COF(Coefficient Of Friction) plot



Force plot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Al.csv
Lab_PMMA_diamond.csv
Lab_Glass_diamond.csv
DAQ.Fx (N)
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0.5
0.0
-0.5
-1.0
0
2
4
6
8
10
12
Timestamp